NANYANG POLYTECHNIC

Course	: Diploma in Electronic Systems (EGDF18) Diploma in Electronic & Computer Engineering (EGDF20)
Module	: Embedded System Design & Technology (EGE320)
Tutorial No	: Tutorial 4
Tutorial Title	: LCD Screen
Objective	: Describe the graphics system hardware components by recognizing the properties and purposes of each component in the system.

Questions:

1. Which of the following item is used by embedded device to provide visual feedback to the users?

   A. LCD Screen
   B. Touch Panel
   C. Push Button
   D. Keypad

2. How does a LCD screen produce random or fix images in color or monochrome?

   A. Using backlight
   B. Using frontlight
   C. Using sunlight
   D. Using fluorescent light

3. Which of the following LCD screen can produce images with true color?

   A. Segment LCD
   B. Dot Matric LCD
   C. Graphic LCD
   D. Memory LCD

4. Each pixel is composed of three basic colors **EXCEPT**

   A. Red
   B. Green
   C. Blue
   D. White

5. A pixel is able to produce 16 million colors. How many bits are used to represent a single color in a pixel assuming all 3 colors are assigned the same number of bits?

    A. 6-bits
    B. 7-bits
    C. 8-bits
    D. 9-bits

6. How many bits are needed by a pixel to be able to produce High Color?

    A. 13-bits
    B. 14-bits
    C. 15-bits
    D. 16-bits

7. Which of the following is a **VALID** color scheme used to represent 3 different colors in a 16-bits pixel.

    A. 6-6-6
    B. 5-6-5
    C. 8-8-8
    D. 7-7-7

8. What is the commonly used measurement unit to measure the diagonal length of a LCD screen?

    A. mm
    B. cm
    C. m
    D. inch

9. Which of the following does **NOT** belong to the graphics system hardware components?

    A. Touch Controller
    B. Display Glass
    C. Display Controller
    D. Frame Buffer

10. Calculate the Frame Buffer size needed for a LCD screen with 240*240 resolution using a 24 BPP (bit per pixel) color depth.

    A. 57600 bytes
    B. 115200 bytes
    C. 153600 bytes
    D. 172800 bytes